

# Skills Summary

## Linear, Squared and Cubed Scale Factors

*Use this reference while completing your worksheet or when studying.*

**One factor, three jobs.** If every *length* of a figure is multiplied by  $k$ , then

$$\text{length} \times k, \quad \text{area} \times k^2, \quad \text{volume} \times k^3.$$

**Why:** an area is a product of two lengths, so it collects  $k$  twice; a volume is a product of three lengths, so it collects  $k$  three times. Count the dimensions and you know the power.

### Skill 1 — Scaling an area

**Rule:** new area = old area  $\times k^2$ . This holds for any shape and for surface area as well as flat area, because both are two-dimensional.

**Example:** A shape of area  $15 \text{ cm}^2$  has every length multiplied by 4. Find the new area.

**Step 1.** The quantity asked for is an area, which is two-dimensional, so the scale factor is applied twice — square it rather than using it as given.

**Step 2.** new area =  $15 \times 4^2 = 15 \times 16$   
 $\Rightarrow$  **240  $\text{cm}^2$**

**Try it:** A shape of area  $7 \text{ cm}^2$  has every length multiplied by 5. Find the new area.

check: 175

### Skill 2 — Scaling a volume or a capacity

**Rule:** new volume = old volume  $\times k^3$ . Capacity (litres, millilitres) is a volume, so it follows the same rule — and so does mass, when the material is unchanged.

**Example:** A solid of volume  $20 \text{ cm}^3$  has every length multiplied by 3. Find the new volume.

**Step 1.** A volume is three-dimensional, so each of its three length factors is stretched — the multiplier is the cube of the scale factor, not the factor itself.

**Step 2.** new volume =  $20 \times 3^3 = 20 \times 27$   
 $\Rightarrow$  **540  $\text{cm}^3$**

**Try it:** A solid of volume  $12 \text{ cm}^3$  has every length multiplied by 2. Find the new volume.

check: 96

### Skill 3 — Working back to the linear factor

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**Rules, run backwards:** given an area ratio,  $k = \sqrt{\text{area ratio}}$ . Given a volume ratio,  $k = \sqrt[3]{\text{volume ratio}}$ .

The ratio you are handed is *already*  $k^2$  or  $k^3$ , so undo the power before using  $k$  for anything else.

**Example:** Two similar shapes have areas 9 and 144. Find the linear scale factor.

**Step 1.** The given ratio is a ratio of areas, so it equals  $k$  squared, not  $k$  — the move is a square root before anything else.

**Step 2.**  $k^2 = 144/9 = 16$ , so  $k = \sqrt{16}$

$$\Rightarrow k = 4$$

**Try it:** Two similar solids have volumes 8 and 216. Find the linear scale factor.

⌘ check